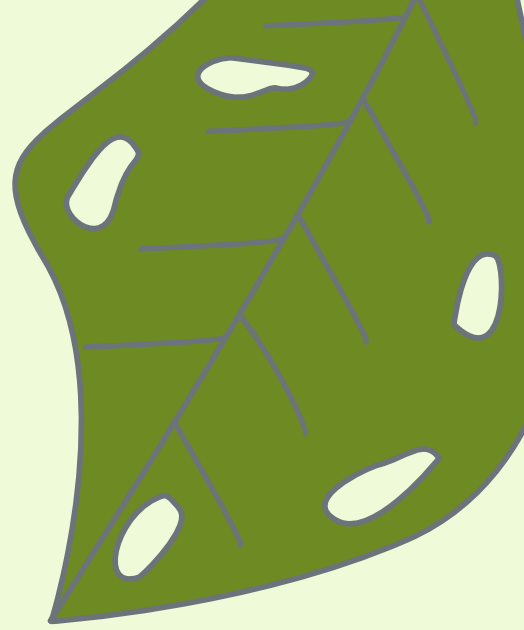
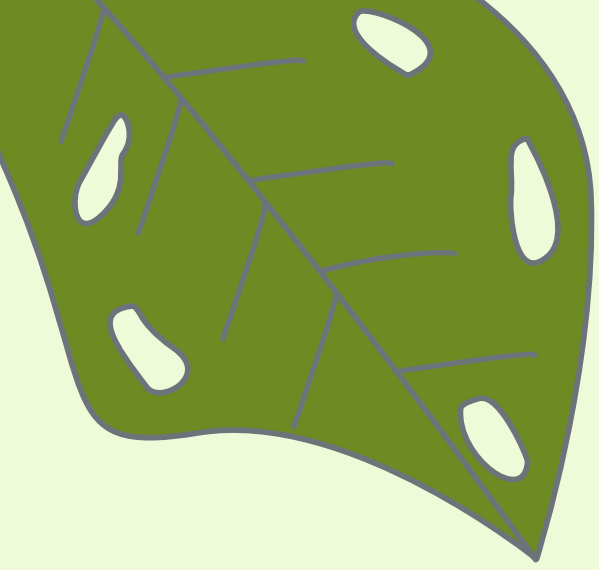
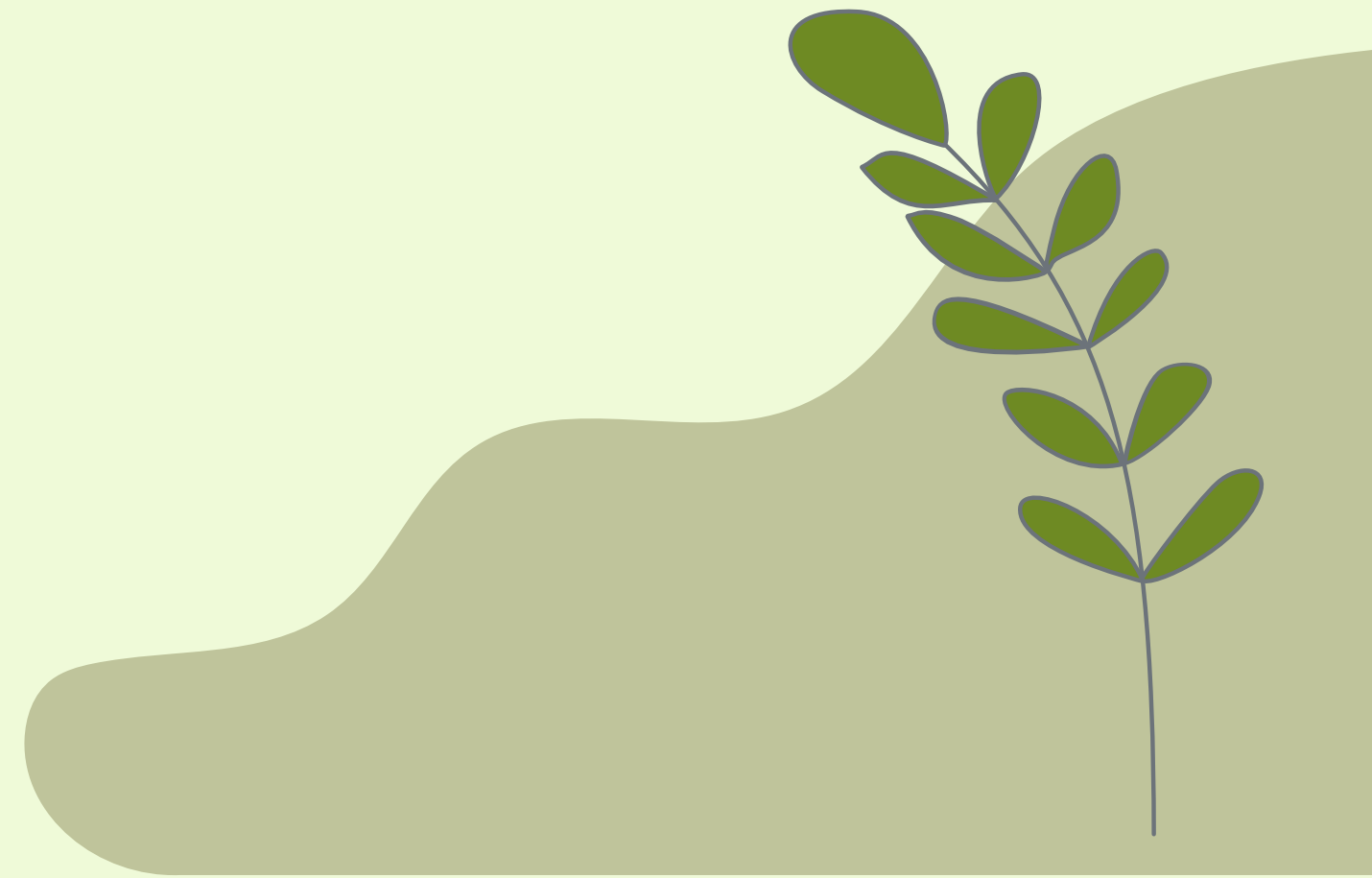
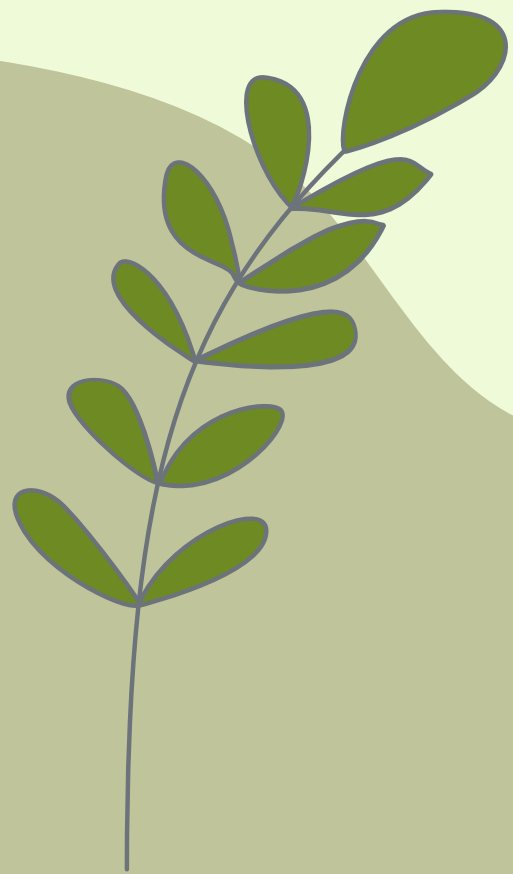




BIOLOGY

in our world

by Jadon Li

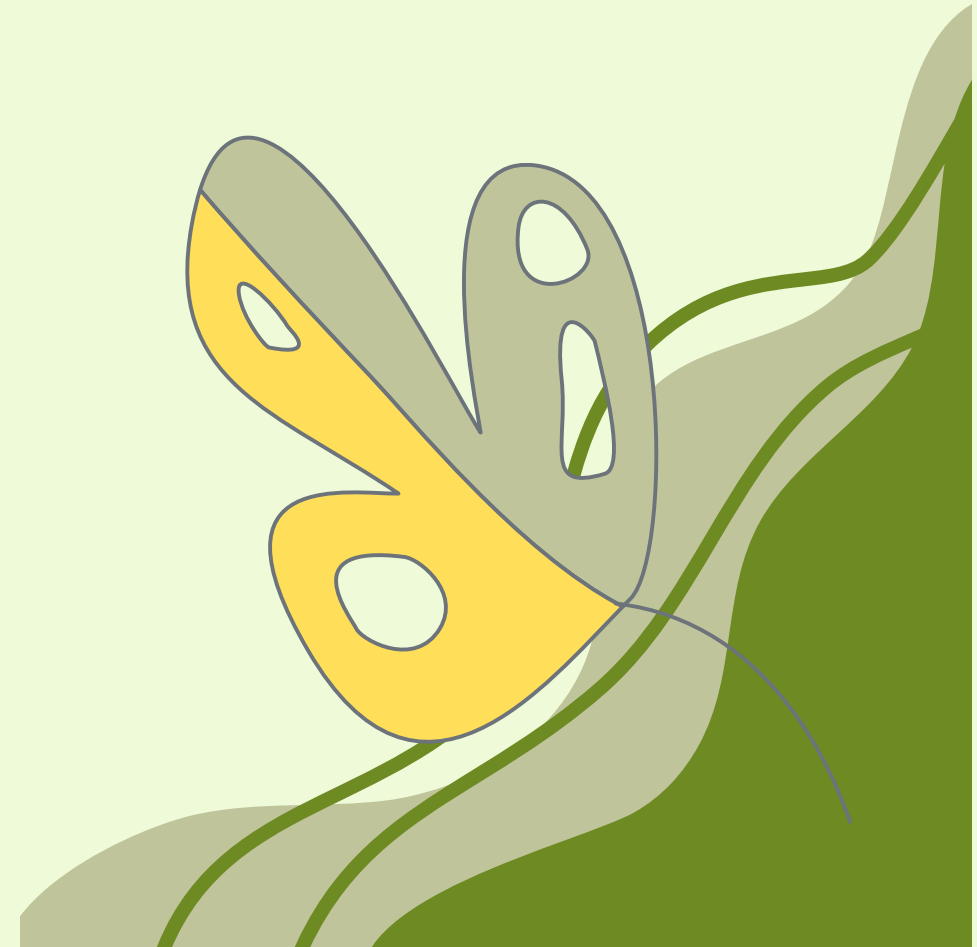


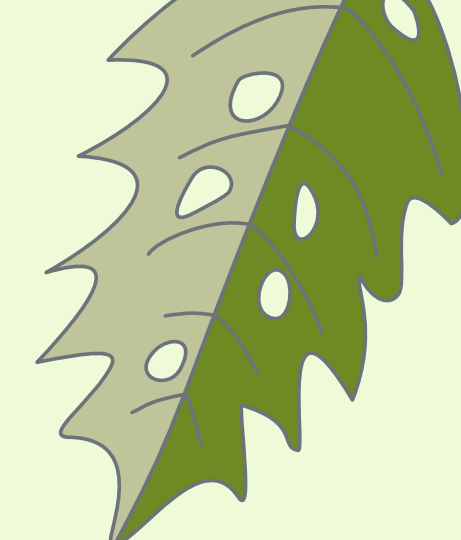
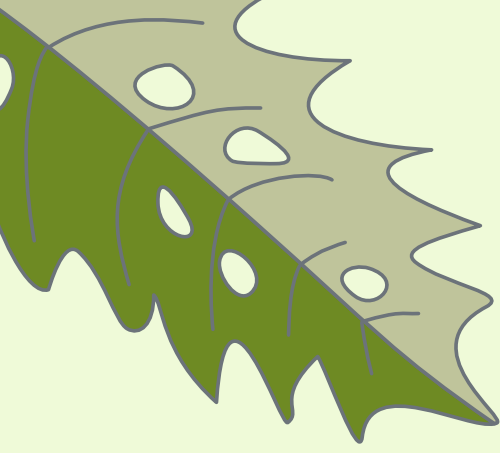
Mentor:



Jadon Li

- AP Biology 5 (Freshmen)
- USBAO Top 15% (Sophomore)
- ACSEF Bioinformatics 3rd (Sophomore)





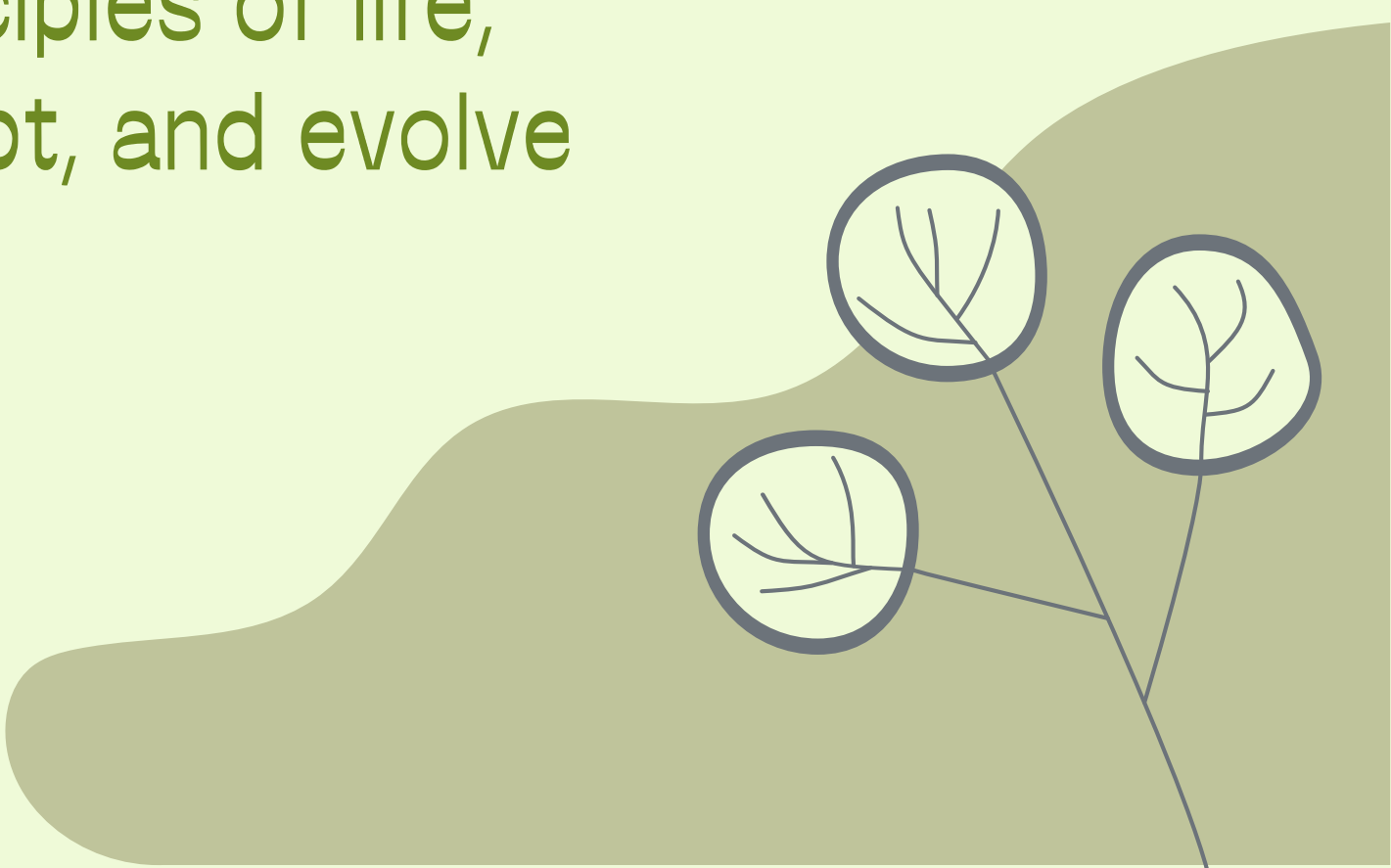
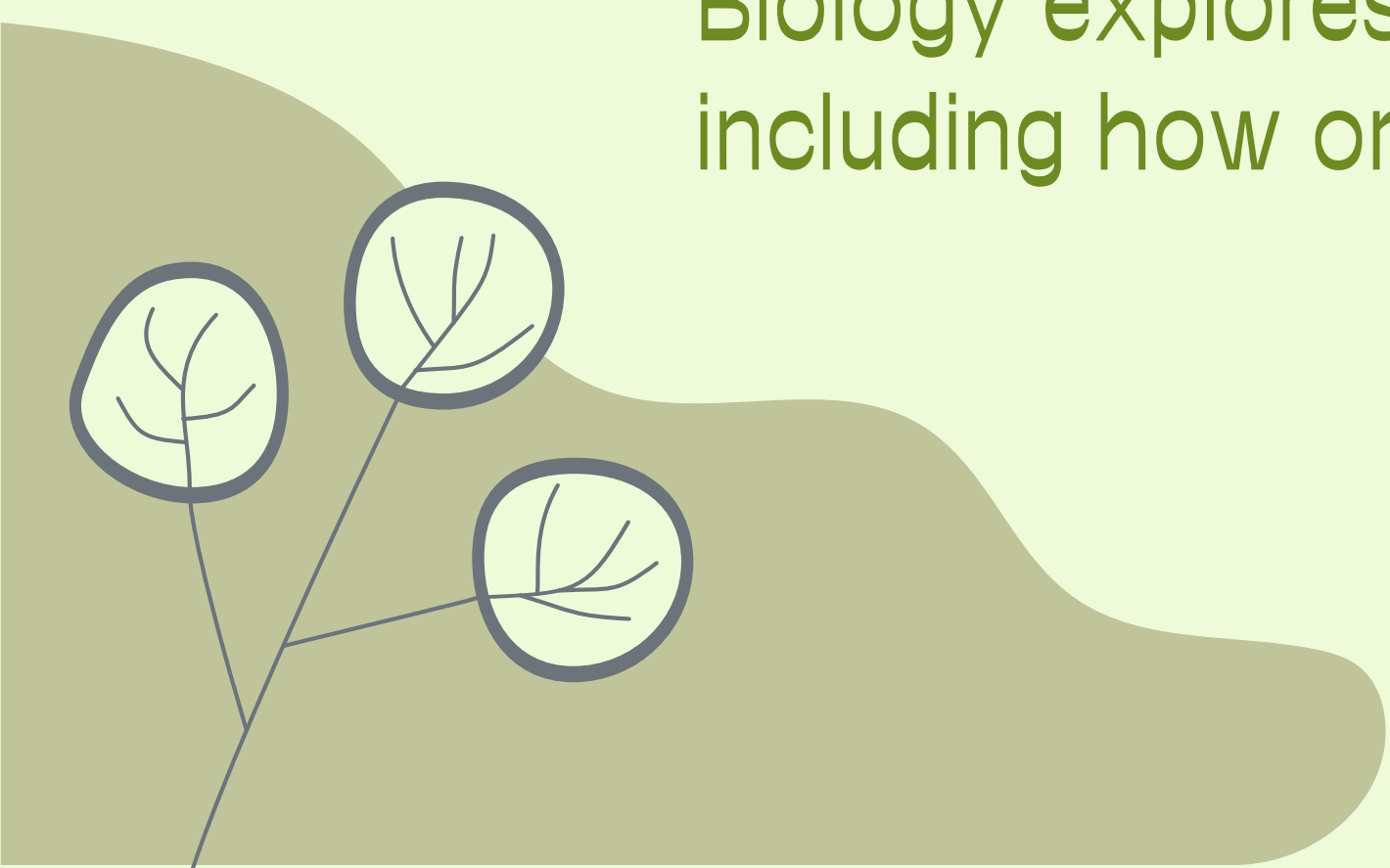
Why Study Biology?

Career Opportunities

a wide range of career paths, including healthcare, research, environmental conservation, education, and more

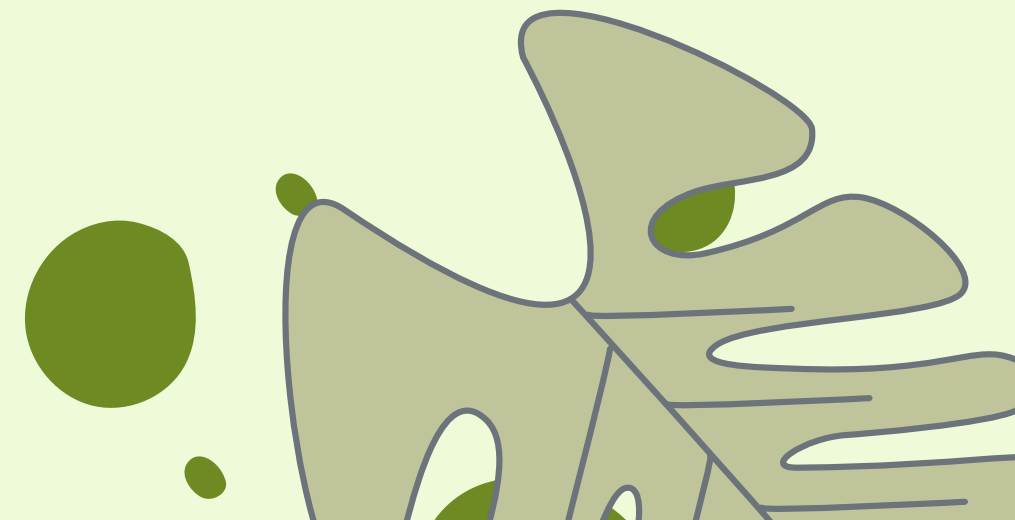
Understanding Life and Its Processes

Biology explores the fundamental principles of life, including how organisms function, adapt, and evolve



Learning Objectives

1. Dissect and annotate primary research papers efficiently.
2. Interpret key experimental methods and data figures.
3. Apply concepts of therapeutic nanobodies and biological resilience across scales.
4. Collaborate to split readings, plan group work, and deliver a polished presentation.
5. Communicate scientific findings clearly, both verbally and visually.



Research Papers

Llama-derived Nanobodies Fight COVID-19 Variants

SARS-CoV-2 Specific Nanobodies Neutralize Different Variants of Concern and Reduce Virus Load in the Brain of h-ACE2 Transgenic Mice

[María Florencia Pavan](#)¹, [Marina Bok](#)^{2,3}, [Rafael Betanzos San Juan](#)⁴, [Juan Pablo Malito](#)^{2,3}, [Gisela Ariana Marcoppido](#)⁵, [Diego Rafael Franco](#)⁵, [Daniela Ayelen Militelo](#)¹, [Juan Manuel Schammas](#)³, [Sara Elizabeth Bari](#)¹, [William Stone](#)⁶, [Krisangel López](#)⁶, [Danielle LaBrie Porier](#)⁶, [John Anthony Muller](#)⁶, [Albert Jonathan Auguste](#)^{6,7}, [Lijuan Yuan](#)^{7,8}, [Andrés Wigdorovitz](#)^{2,3}, [Viviana Gladys Parreño](#)^{2,3,8,*†}, [Lorena Itat Ibañez](#)^{1,*†}

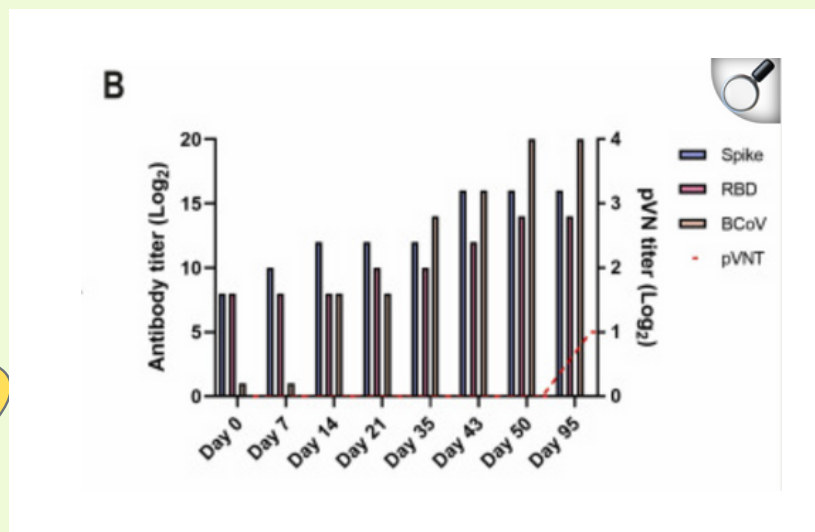
Understanding the Importance of Biological Resilience

Understanding and applying biological resilience, from genes to ecosystems

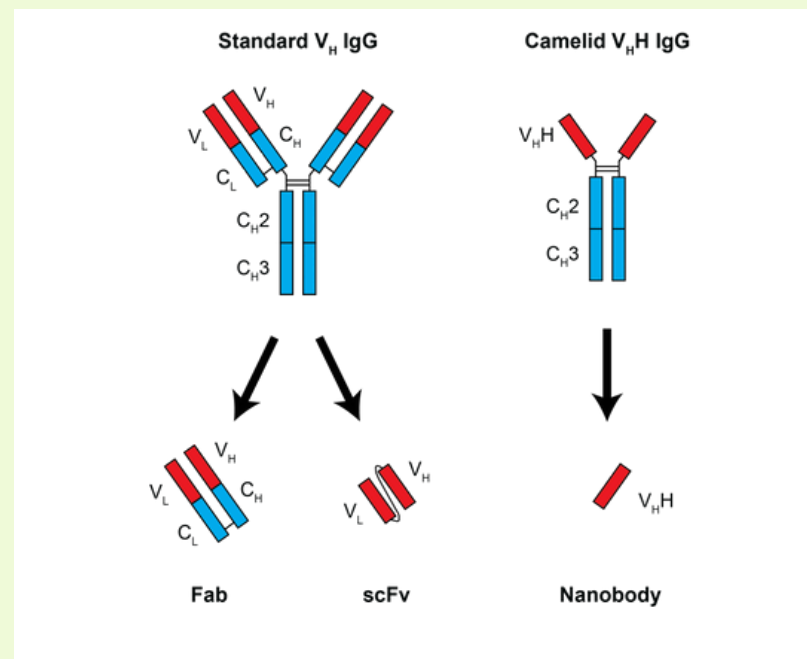
[Rose Thorogood](#) , [Ville Mustonen](#), [Alexandre Aleixo](#), [Pedro J. Aphalo](#), [Fred O. Asiegbu](#), [Mar Cabeza](#), [Johannes Cairns](#), [Ulrika Candolin](#), [Pedro Cardoso](#), [Jussi T. Eronen](#), [Maria Hällfors](#), [Iiris Hovatta](#), [Aino Juslén](#), [Andriy Kovalchuk](#), [Jonna Kulmuni](#), [Liisa Kuula](#), [Raisa Mäkipää](#), [Otso Ovaskainen](#), [Anu-Katriina Pesonen](#), [Craig R. Primmer](#), [Marjo Saastamoinen](#), [Alan H. Schulman](#), [Leif Schulman](#), [Giovanni Strona](#) & [Jarno Vanhatalo](#)

Activities

Interpreting
diagrams and
plots

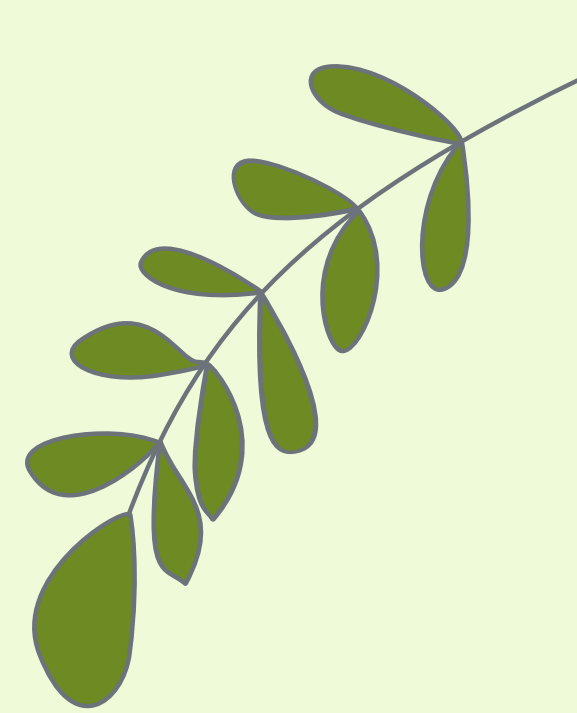
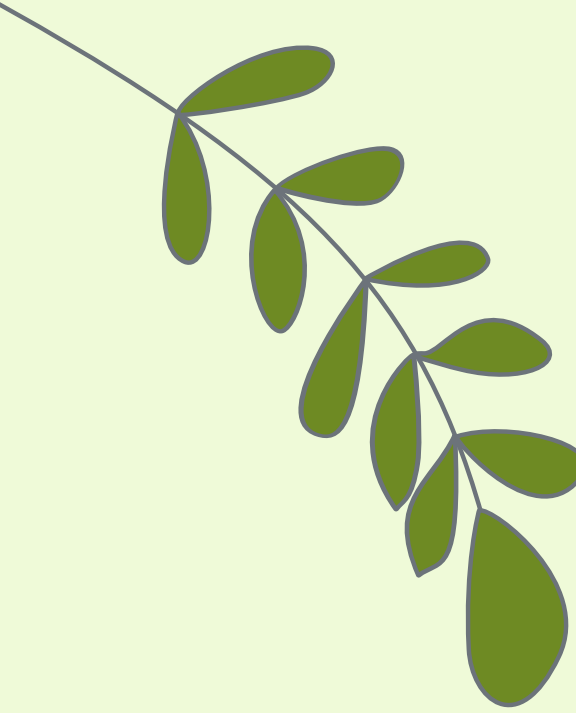


Designing your
own nanobodies



Ecological
Disturbance
escape room





prep for HS Biology
think critically about biology
interpret scientific papers

JOIN BIO!

